Task effects in self-paced reading: A methodological study with Czech readers

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Background: Self-paced reading has been widely used to study sentence processing, yet relatively little is known about how post-reading tasks influence reading-time measures. Although some evidence suggests that task type can affect reading behavior (Swets et al., 2008; Bader & Meng, 2018; Berzak et al., 2025; Chromý & Tomaschek, 2024; Laurinavichyute & von der Malsburg, 2024), systematic investigations specifically are still lacking.

Method: The current study addresses this gap by comparing three moving-window self-paced reading experiments using identical stimuli but different post-reading tasks. Exp1 used yes/no comprehension questions (N=248), Exp2 employed open-ended recall questions (N=258), and Exp3 asked participants for naturalness judgments on a 7-point Likert scale (N=257). We included six experimental types, each with 16 items: (i) Garden-path sentences: garden-path vs. controls; (ii) Plausibility: plausible vs. implausible items; (iii) Ungrammaticality (agreement violations): grammatical vs. ungrammatical; (iv) Gender interference: matched (interfering) gender vs. non-matched (non-interfering) gender; (v) Generic masculine ambiguity: sentences referring to gender-mixed vs. masculine-only referents; (vi) Focus: focused adjuncts vs. unfocused adjuncts. All stimuli were in Czech and adapted from prior research. To ensure task comparability, each experiment included eight control items with explicit instructions (e.g., “Answer ‘yes’ to the following question”). See our OSF entry: https://shorturl.at/nlzkH.

Results & Discussion: See Figure 1 for RTs in the critical and spillover regions for the individual experiment types sorted by task. For each sentence type, we coded one condition to be the affective one, i.e., the one where we expected the effect to surface (GP, implausible, ungrammatical, interfering, gender-mixed, or focused sentences). We coded the other condition as the control one. RTs were analyzed using Bayesian shifted log-normal mixed-effects models with fixed effects Condition*Task and maximal random structures for participants and items. We omit the posterior estimates and other values obtained from our Bayesian models for the sake of brevity in this abstract, but they are present in an analysis *html* file in the OSF repository. In general, across all experimental types, RTs were the fastest in the naturalness judgment experiment, slower in yes/no comprehension, and the slowest in open-ended questions. Task interacted with Condition in selected experimental types. In the ungrammaticality condition (iii), a positive interaction with recall and naturalness judgment comprehension questions indicated a reduced slowdown for simple yes-no questions, likely because this task does not require explicit reflection on grammaticality, unlike the other two tasks. In the interference condition (iv), a negative interaction for naturalness judgments showed that gender-based interference caused greater slowdowns when the task demanded sentence-level recall or answers to yes-no comprehension questions, in contrast to naturalness judgments, arguably because the latter leads to the focus on grammaticality. We also analyzed how the type of task influences sentence wrap-up effects. Here, the relative ordering of reading speed flipped between tasks. While open-ended questions were still the slowest, naturalness judgments became either slower than or equal to yes/no questions. These findings indicate that the depth of sentence processing during self-paced reading is modulated by the nature of the task participants undertake after each sentence. Moreover, the sensitivity to linguistic phenomena varies depending on task demands. This study contributes methodologically to sentence processing research by demonstrating that established effects in self-paced reading can vary as a function of the secondary task, underscoring the importance of task selection in experimental design.

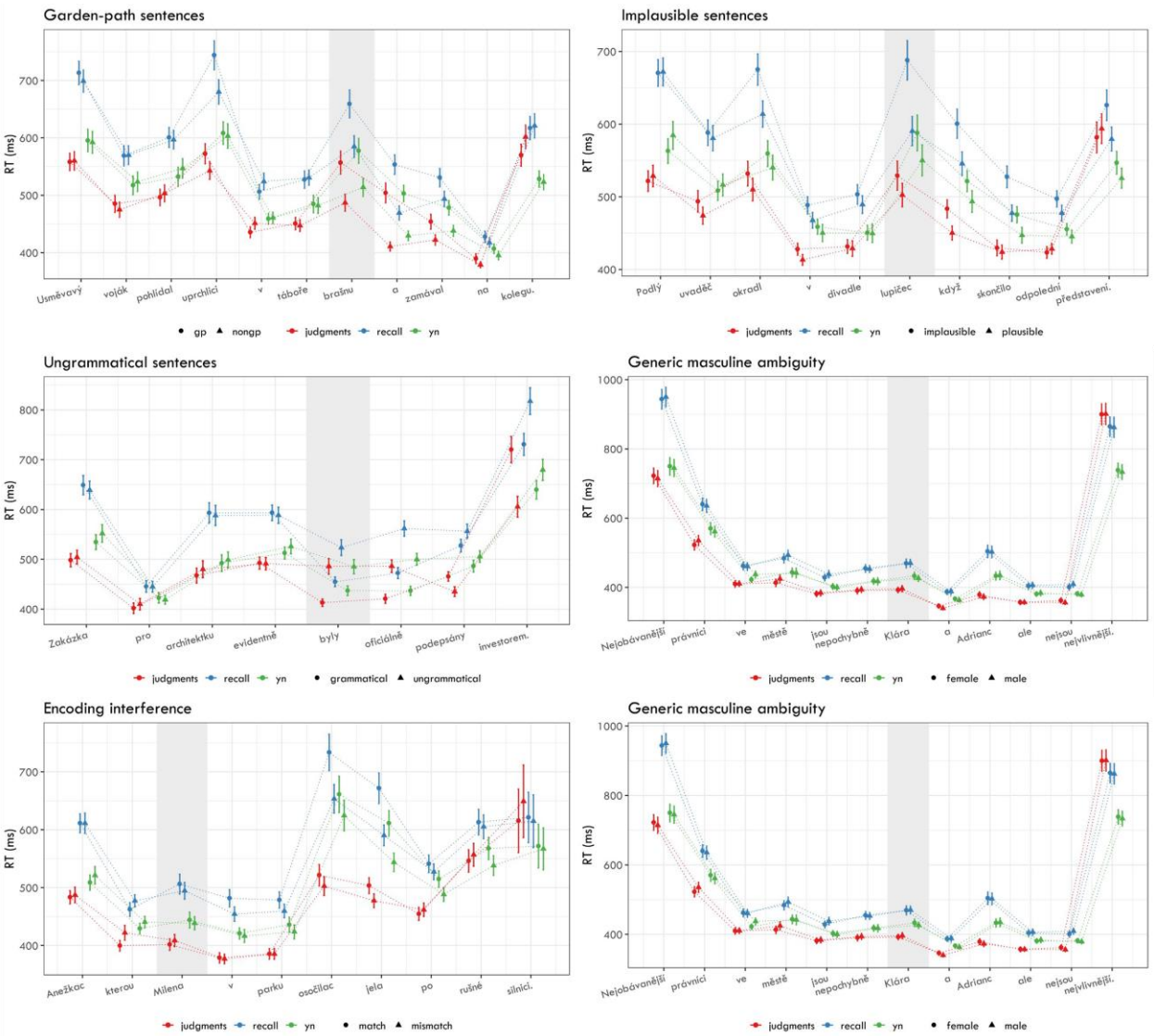

Figure 1: Mean and SEs of RTs per experimental type. Yes-no questions (Exp1) are in green, open-ended ones (Exp2) in blue and naturalness judgments (Exp3) in red.

References: Bader, M., & Meng, M. (2018). The misinterpretation of noncanonical sentences revisited. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 44(8), 1286–1311.; Berzak, Y. et al. (2025). OneStop: A 360-participant english eye tracking dataset with different reading regimes. *Scientific Data*.; Chromý, J., & Tomaschek, F. (2024). Learning or boredom? Task adaptation effects in sentence processing experiments. *Open Mind*, 8, 1447–1468.; Laurinavichyute, A., & Von der Malsburg, T. (2024). Agreement attraction in grammatical sentences and the role of the task. *Journal of Memory and Language*, 137, 104525.; Swets, B., Desmet, T., Clifton, C., & Ferreira, F. (2008). Underspecification of syntactic ambiguities: Evidence from self-paced reading. *Memory & Cognition*, 36, 201–216.